Accesco Hybrid Replenishment Engine — Local PostgreSQL API Test Execution Report

Execution Date: 2026-11-10
Environment: Local Docker PostgreSQL 16 (`accesco_postgres_local` on `localhost:5432/accesco_local`) + FastAPI `2.1.0` (`asyncpg` + SQLAlchemy Async)
Test Suite: `test_case/test_api.py` (`pytest` 9.2.0 + `httpx` + `pytest-asyncio`)
Overall Result: 5 / 5 Edge-Case Tests PASSED (100%) + 10 / 10 Live HTTP End-to-End Checks PASSED

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1. Local PostgreSQL Database Connection Status

- **Provisioning Method:** Docker Compose (`docker-compose.yml` running `postgres:16-alpine` container `accesco_postgres_local`)
- **Connection String (`.env`):** `postgresql+asyncpg://accesco:accesco_local_pass@localhost:5432/accesco_local`
- **Container Health Status:** `Up (healthy)` on `0.0.0.0:5432->5432/tcp`
- **Schema & Seed Application:** Applied `sql/01_schema_ddl.sql` and `sql/02_seed_dml.sql` via `sql/apply_schema.py`
- **Verified PostgreSQL Tables (10 Active Tables):**
  1. `cdao_replenishment_priors` (Phase 1 SKU catalog, category buffers, 1p/2p catchment priors, MOQ, case pack, `doorstep_rejection_factor_r`)
  2. `wms_inventory_batch` (Phase 1 batch tracking with `harvest_date`, `manufacture_date`, `expiry_date`, `status`, `hours_since_harvest`, `life_fraction_l`)
  3. `oms_recipe_bom` (Phase 1 meal-to-ingredient Bill of Materials)
  4. `cdao_calendar_factors` (Phase 1 dynamic `WEEKDAY` and `MONTH_CYCLE` multipliers)
  5. `oms_households` (Household profiles, `days_since_active`, `is_opted_out`, `in_active_cart`, cooldown tracking)
  6. `oms_customer_purchase_log` (Customer purchase event log)
  7. `cdao_kitchen_capacity` (Category-specific & aggregate Swadishtt kitchen absorption limits `K`)
  8. `cdao_system_config` (Kill-switch toggle, quiet hours, cooldown days, frequency cap, latest `E_h` bottom-up demand)
  9. `cdao_po_proposals` (Store pull PO proposals, alerts, and warehouse approval workflow states)
  10. `cdao_nudge_audit_log` (Immutable evaluation and suppression audit records)

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2. Executed Test Cases Summary

Test ID	Test Name	Target Endpoint(s)	Edge Case Validated	Status
`test_00`	`test_00_postgres_healthcheck`	`GET /health`	Verifies PostgreSQL `asyncpg` connectivity and	**PASSED**

Test ID	Test Name	Target Endpoint(s)	Edge Case Validated	Status
			kitchen capacity caps loaded from DB	
`test_01`	`test_01_seven_day_dormancy_rule`	`POST /jobs/hce-nudge`	Household `TEST-HH-DORMANT-8D` inactive for 8 days (`days_since_active = 8`) is suppressed with `BLOCKED_DORMANT` and logged in `cdao_nudge_audit_log`	**PASSED**
`test_02`	`test_02_doorstep_rejection_decoupling`	`POST /jobs/store-pull`	Setting `doorstep_rejection_factor_r = 0.650` on `SKU-TOMATO-HYBRID` triggers a `HIGH PROCUREMENT_ALERT` without reducing `reconciled_demand_D` (`53.35`)	**PASSED**
`test_03`	`test_03_kitchen_capacity_category_splits`	`POST /jobs/store-pull`	Massive fresh overflow trims buffers strictly in order `leafy_greens` (10%→0%) → `dairy` (10%→0%) → `base_veg` (15%→6%), keeping overflow (`18.04 kg`) under caps	**PASSED**
`test_04`	`test_04_end_to_end_aging_exclusion_flow`	`POST /jobs/fresh-routing` → `POST /jobs/store-pull`	52h-old `SKU-PANEER-FRESH` batch (10 units) routes to `ROUTE_TO_KITCHEN` in PostgreSQL and is excluded from `sound_on_hand` (`6.0` units only) in `/jobs/store-pull`	**PASSED**

3. Exact Request & Response Payloads

Test 0: PostgreSQL Healthcheck (`test_00_postgres_healthcheck`)

Request Sent:

```
{
  "method": "GET",
  "url": "/health",
  "query": null,
  "body": null
}
```


Response Received (`HTTP 200`):


```
{
  "service": "Accesco Hybrid Replenishment Engine",
  "version": "2.1.0",
  "status": "HEALTHY",
  "storage_mode": "PostgreSQL (asyncpg)",
  "database_connected": true,
  "kill_switch_active": false,
  "kitchen_capacity_caps_kg": {
    "leafy_greens": 8.0,
    "dairy": 12.0,
    "base_veg": 20.0,
    "total_fresh": 35.0
  },
  "cooldown_days": 3,
  "quiet_hours": "21:00 to 8:00",
  "timestamp": "2026-11-10T15:08:55.102514"
}
```

Test 1: The 7-Day Dormancy Rule (`test_01_seven_day_dormancy_rule`)

Precondition Inserted into PostgreSQL:

- `oms_households`: `household_id = 'TEST-HH-DORMANT-8D'`, `is_active = TRUE`, `days_since_active = 8`
- `oms_customer_purchase_log`: `SKU-MILK-1L`, `quantity = 2.000`, `purchase_date = NOW() - INTERVAL '10 days'`

Request Sent:


```
{
  "method": "POST",
  "url": "/jobs/hce-nudge?run_sync=true&ignore_quiet_hours=true",
  "query": {
    "run_sync": true,
    "ignore_quiet_hours": true
  },
  "body": null
}
```


Response Received (`HTTP 200`):


```
{
  "timestamp": "2026-11-10T09:38:55.258720Z",
  "kill_switch_active": false,
  "total_evaluations": 8,
  "nudges_generated": 2,
```

```
  "nudges_blocked": 6,
  "nudge_details": [
    {
      "household_id": "H101",
      "sku_id": "SKU-MILK-1L",
      "sku_name": "Fresh Farm Milk (1 Litre)",
      "purchases_count": 4,
      "shrunk_rate_r_hi": 1.42,
      "days_since_last_purchase": 1,
      "expected_low_stock_days": 0.1,
      "reorder_probability": 0.094,
      "nudge_eligible": false,
      "gate_status": "PASSED",
      "reason": "All eligibility, stock, consent, and policy criteria passed."
    },
    {
      "household_id": "H101",
      "sku_id": "SKU-SPINACH-BUNCH",
      "sku_name": "Farm Fresh Palak / Spinach (250g)",
      "purchases_count": 2,
      "shrunk_rate_r_hi": 0.3667,
      "days_since_last_purchase": 4,
      "expected_low_stock_days": -1.8,
      "reorder_probability": 0.5399,
      "nudge_eligible": true,
      "gate_status": "PASSED",
      "reason": "All eligibility, stock, consent, and policy criteria passed."
    },
    {
      "household_id": "H102",
      "sku_id": "SKU-MILK-1L",
      "sku_name": "Fresh Farm Milk (1 Litre)",
      "purchases_count": 1,
      "shrunk_rate_r_hi": 1.0,
      "days_since_last_purchase": 2,
      "expected_low_stock_days": -1.2,
      "reorder_probability": 0.845,
      "nudge_eligible": true,
      "gate_status": "PASSED",
      "reason": "All eligibility, stock, consent, and policy criteria passed."
    },
    {
      "household_id": "H103",
      "sku_id": "SKU-MILK-1L",
      "sku_name": "Fresh Farm Milk (1 Litre)",
      "purchases_count": 1,
      "shrunk_rate_r_hi": 1.0,
      "days_since_last_purchase": 15,
      "expected_low_stock_days": -14.2,
      "reorder_probability": 0.0,
      "nudge_eligible": false,
      "gate_status": "BLOCKED_DORMANT",
      "reason": "Customer dormant (12 days inactive > 7d threshold)."
    },
    {
      "household_id": "H104",
      "sku_id": "SKU-SPINACH-BUNCH",
      "sku_name": "Farm Fresh Palak / Spinach (250g)",
```

```
      "purchases_count": 1,
      "shrunk_rate_r_hi": 0.25,
      "days_since_last_purchase": 4,
      "expected_low_stock_days": -0.8,
      "reorder_probability": 0.5399,
      "nudge_eligible": false,
      "gate_status": "BLOCKED_OPT_OUT",
      "reason": "Customer has opted out of replenishment reminders."
    },
    {
      "household_id": "H105",
      "sku_id": "SKU-TOMATO-HYBRID",
      "sku_name": "Country Tomatoes (1kg)",
      "purchases_count": 1,
      "shrunk_rate_r_hi": 0.25,
      "days_since_last_purchase": 5,
      "expected_low_stock_days": -1.8,
      "reorder_probability": 0.8081,
      "nudge_eligible": false,
      "gate_status": "BLOCKED_IN_CART",
      "reason": "Item SKU-TOMATO-HYBRID is already in customer's active shopping cart."
    },
    {
      "household_id": "H106",
      "sku_id": "SKU-MILK-1L",
      "sku_name": "Fresh Farm Milk (1 Litre)",
      "purchases_count": 1,
      "shrunk_rate_r_hi": 1.0,
      "days_since_last_purchase": 3,
      "expected_low_stock_days": -2.2,
      "reorder_probability": 0.9929,
      "nudge_eligible": false,
      "gate_status": "BLOCKED_COOLDOWN",
      "reason": "Household is in cooldown (min 3 days between reminders)."
    },
    {
      "household_id": "TEST-HH-DORMANT-8D",
      "sku_id": "SKU-MILK-1L",
      "sku_name": "Fresh Farm Milk (1 Litre)",
      "purchases_count": 1,
      "shrunk_rate_r_hi": 1.0,
      "days_since_last_purchase": 10,
      "expected_low_stock_days": -8.4,
      "reorder_probability": 0.0,
      "nudge_eligible": false,
      "gate_status": "BLOCKED_DORMANT",
      "reason": "Customer dormant (8 days inactive > 7d threshold)."
    }
  ],
  "bottom_up_demand_Eh": {
    "SKU-ATTA-5KG": 0.0,
    "SKU-CORIANDER-100G": 0.0,
    "SKU-MILK-1L": 3.86,
    "SKU-PANEER-FRESH": 0.0,
    "SKU-POTATO-1KG": 0.0,
    "SKU-SPINACH-BUNCH": 1.08,
    "SKU-TOMATO-HYBRID": 0.81,
    "SKU-TOOR-DAL-1KG": 0.0
```

```
  }
}
```

Test 2: Doorstep Rejection Decoupling (`test_02_doorstep_rejection_decoupling`)

Precondition Updated in PostgreSQL:

- `cdao_replenishment_priors`: `SKU-TOMATO-HYBRID` updated to `doorstep_rejection_factor_r = 0.650` (baseline `D = 53.35` when `R = 1.000`).

Request Sent:


```
{
  "method": "POST",
  "url": "/jobs/store-pull?run_sync=true&delivery_date=2026-11-11",
  "precondition": {
    "sku_id": "SKU-TOMATO-HYBRID",
    "doorstep_rejection_factor_r": 0.65,
    "baseline_D": 53.35
  },
  "body": null
}
```


Response Received (`HTTP 200`):


```
{
  "po_id": "PO-20261111-5910A7",
  "created_at": "2026-11-10T09:38:57.522057Z",
  "status": "PENDING_APPROVAL",
  "target_delivery_date": "2026-11-11",
  "festival_factor_F": 1.0,
  "month_salary_factor_M": 1.0,
  "doorstep_rejection_factor_R": 0.65,
  "kitchen_capacity_cap_kg": 35.0,
  "category_kitchen_caps_kg": {
    "leafy_greens": 8.0,
    "dairy": 12.0,
    "base_veg": 20.0,
    "total_fresh": 35.0
  },
  "category_overflow_kg": {
    "leafy_greens": 6.72,
    "dairy": 8.54,
    "base_veg": 19.64
  },
  "total_fresh_overflow_kg": 34.9,
  "overflow_cap_exceeded": true,
  "buffer_reduction_iterations": 1,
  "total_proposed_units": 220,
  "procurement_alerts": [
    {
      "sku_id": "SKU-TOMATO-HYBRID",
      "sku_name": "Country Tomatoes (1kg)",
      "rejection_factor_r": 0.65,
```

```
      "severity": "HIGH",
      "recommended_action": "PROCUREMENT_ALERT: Doorstep rejection factor R=0.65 below 0.95 threshold. Audit f
    }
  ],
  "items": [
    {
      "sku_id": "SKU-ATTA-5KG",
      "name": "Chakki Fresh Sharbati Atta (5kg)",
      "category": "fmcg",
      "trailing_avg_Aiw": 18.0,
      "bottom_up_Eh": 0.0,
      "divergence_pct": 0.0,
      "base_demand": 18.0,
      "day_constant_c": 0.97,
      "month_factor_M": 1.0,
      "rejection_factor_R": 1.0,
      "reconciled_demand_D": 17.46,
      "initial_buffer_percentage": 25.0,
      "buffer_percentage_applied": 25.0,
      "order_up_to_S": 21.83,
      "sound_on_hand": 17.0,
      "raw_order": 4.83,
      "case_pack_size": 4,
      "moq": 8,
      "proposed_quantity": 8,
      "approved_quantity": null,
      "kitchen_overflow_kg": 0.0
    },
    {
      "sku_id": "SKU-CORIANDER-100G",
      "name": "Fresh Coriander Leaves (100g)",
      "category": "leafy_greens",
      "trailing_avg_Aiw": 45.0,
      "bottom_up_Eh": 0.0,
      "divergence_pct": 0.0,
      "base_demand": 45.0,
      "day_constant_c": 0.97,
      "month_factor_M": 1.0,
      "rejection_factor_R": 1.0,
      "reconciled_demand_D": 43.65,
      "initial_buffer_percentage": 10.0,
      "buffer_percentage_applied": 9.0,
      "order_up_to_S": 47.58,
      "sound_on_hand": 0.0,
      "raw_order": 47.58,
      "case_pack_size": 10,
      "moq": 20,
      "proposed_quantity": 50,
      "approved_quantity": null,
      "kitchen_overflow_kg": 3.93
    },
    {
      "sku_id": "SKU-MILK-1L",
      "name": "Fresh Farm Milk (1 Litre)",
      "category": "dairy",
      "trailing_avg_Aiw": 70.0,
      "bottom_up_Eh": 0.0,
      "divergence_pct": 0.0,
```

```
      "base_demand": 70.0,
      "day_constant_c": 0.97,
      "month_factor_M": 1.0,
      "rejection_factor_R": 1.0,
      "reconciled_demand_D": 67.9,
      "initial_buffer_percentage": 10.0,
      "buffer_percentage_applied": 10.0,
      "order_up_to_S": 74.69,
      "sound_on_hand": 39.0,
      "raw_order": 35.69,
      "case_pack_size": 12,
      "moq": 24,
      "proposed_quantity": 36,
      "approved_quantity": null,
      "kitchen_overflow_kg": 6.79
    },
    {
      "sku_id": "SKU-PANEER-FRESH",
      "name": "Artisanal Fresh Paneer (200g)",
      "category": "dairy",
      "trailing_avg_Aiw": 18.0,
      "bottom_up_Eh": 0.0,
      "divergence_pct": 0.0,
      "base_demand": 18.0,
      "day_constant_c": 0.97,
      "month_factor_M": 1.0,
      "rejection_factor_R": 1.0,
      "reconciled_demand_D": 17.46,
      "initial_buffer_percentage": 10.0,
      "buffer_percentage_applied": 10.0,
      "order_up_to_S": 19.21,
      "sound_on_hand": 6.0,
      "raw_order": 13.21,
      "case_pack_size": 1,
      "moq": 5,
      "proposed_quantity": 14,
      "approved_quantity": null,
      "kitchen_overflow_kg": 1.75
    },
    {
      "sku_id": "SKU-POTATO-1KG",
      "name": "Agra Potatoes (1kg)",
      "category": "base_veg",
      "trailing_avg_Aiw": 80.0,
      "bottom_up_Eh": 0.0,
      "divergence_pct": 0.0,
      "base_demand": 80.0,
      "day_constant_c": 0.97,
      "month_factor_M": 1.0,
      "rejection_factor_R": 1.0,
      "reconciled_demand_D": 77.6,
      "initial_buffer_percentage": 15.0,
      "buffer_percentage_applied": 15.0,
      "order_up_to_S": 89.24,
      "sound_on_hand": 40.0,
      "raw_order": 49.24,
      "case_pack_size": 10,
      "moq": 40,
```

```
    "proposed_quantity": 50,
    "approved_quantity": null,
    "kitchen_overflow_kg": 11.64
  },
  {
    "sku_id": "SKU-SPINACH-BUNCH",
    "name": "Farm Fresh Palak / Spinach (250g)",
    "category": "leafy_greens",
    "trailing_avg_Aiw": 32.0,
    "bottom_up_Eh": 0.0,
    "divergence_pct": 0.0,
    "base_demand": 32.0,
    "day_constant_c": 0.97,
    "month_factor_M": 1.0,
    "rejection_factor_R": 1.0,
    "reconciled_demand_D": 31.04,
    "initial_buffer_percentage": 10.0,
    "buffer_percentage_applied": 9.0,
    "order_up_to_S": 33.83,
    "sound_on_hand": 26.0,
    "raw_order": 7.83,
    "case_pack_size": 5,
    "moq": 10,
    "proposed_quantity": 10,
    "approved_quantity": null,
    "kitchen_overflow_kg": 2.79
  },
  {
    "sku_id": "SKU-TOMATO-HYBRID",
    "name": "Country Tomatoes (1kg)",
    "category": "base_veg",
    "trailing_avg_Aiw": 55.0,
    "bottom_up_Eh": 0.0,
    "divergence_pct": 0.0,
    "base_demand": 55.0,
    "day_constant_c": 0.97,
    "month_factor_M": 1.0,
    "rejection_factor_R": 0.65,
    "reconciled_demand_D": 53.35,
    "initial_buffer_percentage": 15.0,
    "buffer_percentage_applied": 15.0,
    "order_up_to_S": 61.35,
    "sound_on_hand": 25.0,
    "raw_order": 36.35,
    "case_pack_size": 10,
    "moq": 30,
    "proposed_quantity": 40,
    "approved_quantity": null,
    "kitchen_overflow_kg": 8.0
  },
  {
    "sku_id": "SKU-TOOR-DAL-1KG",
    "name": "Unpolished Toor Dal (1kg)",
    "category": "fmcg",
    "trailing_avg_Aiw": 22.0,
    "bottom_up_Eh": 0.0,
    "divergence_pct": 0.0,
    "base_demand": 22.0,
```

```
      "day_constant_c": 0.97,
      "month_factor_M": 1.0,
      "rejection_factor_R": 1.0,
      "reconciled_demand_D": 21.34,
      "initial_buffer_percentage": 25.0,
      "buffer_percentage_applied": 25.0,
      "order_up_to_S": 26.68,
      "sound_on_hand": 15.0,
      "raw_order": 11.68,
      "case_pack_size": 6,
      "moq": 12,
      "proposed_quantity": 12,
      "approved_quantity": null,
      "kitchen_overflow_kg": 0.0
    }
  ],
  "reviewed_by": null,
  "reviewed_at": null,
  "override_reason": null
}
```

Test 3: Kitchen Capacity (K) Category Splits (`test_03_kitchen_capacity_category_splits`)

Request Sent:


```
{
  "method": "POST",
  "url": "/jobs/store-pull?run_sync=true&delivery_date=2026-11-11",
  "precondition": {
    "SKU-SPINACH-BUNCH": "demand~200, buffer=10%",
    "SKU-MILK-1L": "demand~250, buffer=10%",
    "SKU-TOMATO-HYBRID": "demand~300, buffer=15%",
    "caps": {
      "leafy_greens": 8.0,
      "dairy": 12.0,
      "base_veg": 20.0,
      "total_fresh": 35.0
    }
  },
  "body": null
}
```


Response Received (`HTTP 200`):


```
{
  "po_id": "PO-20261111-903812",
  "created_at": "2026-11-10T09:38:58.484671Z",
  "status": "PENDING_APPROVAL",
  "target_delivery_date": "2026-11-11",
  "festival_factor_F": 1.0,
  "month_salary_factor_M": 1.0,
  "doorstep_rejection_factor_R": 1.0,
  "kitchen_capacity_cap_kg": 35.0,
```

```
  "category_kitchen_caps_kg": {
    "leafy_greens": 8.0,
    "dairy": 12.0,
    "base_veg": 20.0,
    "total_fresh": 35.0
  },
  "category_overflow_kg": {
    "leafy_greens": 0.0,
    "dairy": 0.0,
    "base_veg": 18.04
  },
  "total_fresh_overflow_kg": 18.04,
  "overflow_cap_exceeded": true,
  "buffer_reduction_iterations": 29,
  "total_proposed_units": 881,
  "procurement_alerts": [],
  "items": [
    {
      "sku_id": "SKU-ATTA-5KG",
      "name": "Chakki Fresh Sharbati Atta (5kg)",
      "category": "fmcg",
      "trailing_avg_Aiw": 18.0,
      "bottom_up_Eh": 0.0,
      "divergence_pct": 0.0,
      "base_demand": 18.0,
      "day_constant_c": 0.97,
      "month_factor_M": 1.0,
      "rejection_factor_R": 1.0,
      "reconciled_demand_D": 17.46,
      "initial_buffer_percentage": 25.0,
      "buffer_percentage_applied": 25.0,
      "order_up_to_S": 21.83,
      "sound_on_hand": 0.0,
      "raw_order": 21.83,
      "case_pack_size": 4,
      "moq": 8,
      "proposed_quantity": 24,
      "approved_quantity": null,
      "kitchen_overflow_kg": 0.0
    },
    {
      "sku_id": "SKU-CORIANDER-100G",
      "name": "Fresh Coriander Leaves (100g)",
      "category": "leafy_greens",
      "trailing_avg_Aiw": 10.0,
      "bottom_up_Eh": 0.0,
      "divergence_pct": 0.0,
      "base_demand": 10.0,
      "day_constant_c": 0.97,
      "month_factor_M": 1.0,
      "rejection_factor_R": 1.0,
      "reconciled_demand_D": 9.7,
      "initial_buffer_percentage": 10.0,
      "buffer_percentage_applied": 0.0,
      "order_up_to_S": 9.7,
      "sound_on_hand": 0.0,
      "raw_order": 9.7,
      "case_pack_size": 10,
```

```
      "moq": 20,
      "proposed_quantity": 20,
      "approved_quantity": null,
      "kitchen_overflow_kg": 0.0
    },
    {
      "sku_id": "SKU-MILK-1L",
      "name": "Fresh Farm Milk (1 Litre)",
      "category": "dairy",
      "trailing_avg_Aiw": 250.0,
      "bottom_up_Eh": 0.0,
      "divergence_pct": 0.0,
      "base_demand": 250.0,
      "day_constant_c": 0.97,
      "month_factor_M": 1.0,
      "rejection_factor_R": 1.0,
      "reconciled_demand_D": 242.5,
      "initial_buffer_percentage": 10.0,
      "buffer_percentage_applied": 0.0,
      "order_up_to_S": 242.5,
      "sound_on_hand": 0.0,
      "raw_order": 242.5,
      "case_pack_size": 12,
      "moq": 24,
      "proposed_quantity": 252,
      "approved_quantity": null,
      "kitchen_overflow_kg": 0.0
    },
    {
      "sku_id": "SKU-PANEER-FRESH",
      "name": "Artisanal Fresh Paneer (200g)",
      "category": "dairy",
      "trailing_avg_Aiw": 10.0,
      "bottom_up_Eh": 0.0,
      "divergence_pct": 0.0,
      "base_demand": 10.0,
      "day_constant_c": 0.97,
      "month_factor_M": 1.0,
      "rejection_factor_R": 1.0,
      "reconciled_demand_D": 9.7,
      "initial_buffer_percentage": 10.0,
      "buffer_percentage_applied": 0.0,
      "order_up_to_S": 9.7,
      "sound_on_hand": 0.0,
      "raw_order": 9.7,
      "case_pack_size": 1,
      "moq": 5,
      "proposed_quantity": 10,
      "approved_quantity": null,
      "kitchen_overflow_kg": 0.0
    },
    {
      "sku_id": "SKU-POTATO-1KG",
      "name": "Agra Potatoes (1kg)",
      "category": "base_veg",
      "trailing_avg_Aiw": 10.0,
      "bottom_up_Eh": 0.0,
      "divergence_pct": 0.0,
```

```
      "base_demand": 10.0,
      "day_constant_c": 0.97,
      "month_factor_M": 1.0,
      "rejection_factor_R": 1.0,
      "reconciled_demand_D": 9.7,
      "initial_buffer_percentage": 15.0,
      "buffer_percentage_applied": 6.0,
      "order_up_to_S": 10.28,
      "sound_on_hand": 0.0,
      "raw_order": 10.28,
      "case_pack_size": 10,
      "moq": 40,
      "proposed_quantity": 40,
      "approved_quantity": null,
      "kitchen_overflow_kg": 0.58
    },
    {
      "sku_id": "SKU-SPINACH-BUNCH",
      "name": "Farm Fresh Palak / Spinach (250g)",
      "category": "leafy_greens",
      "trailing_avg_Aiw": 200.0,
      "bottom_up_Eh": 0.0,
      "divergence_pct": 0.0,
      "base_demand": 200.0,
      "day_constant_c": 0.97,
      "month_factor_M": 1.0,
      "rejection_factor_R": 1.0,
      "reconciled_demand_D": 194.0,
      "initial_buffer_percentage": 10.0,
      "buffer_percentage_applied": 0.0,
      "order_up_to_S": 194.0,
      "sound_on_hand": 0.0,
      "raw_order": 194.0,
      "case_pack_size": 5,
      "moq": 10,
      "proposed_quantity": 195,
      "approved_quantity": null,
      "kitchen_overflow_kg": 0.0
    },
    {
      "sku_id": "SKU-TOMATO-HYBRID",
      "name": "Country Tomatoes (1kg)",
      "category": "base_veg",
      "trailing_avg_Aiw": 300.0,
      "bottom_up_Eh": 0.0,
      "divergence_pct": 0.0,
      "base_demand": 300.0,
      "day_constant_c": 0.97,
      "month_factor_M": 1.0,
      "rejection_factor_R": 1.0,
      "reconciled_demand_D": 291.0,
      "initial_buffer_percentage": 15.0,
      "buffer_percentage_applied": 6.0,
      "order_up_to_S": 308.46,
      "sound_on_hand": 0.0,
      "raw_order": 308.46,
      "case_pack_size": 10,
      "moq": 30,
```

```
      "proposed_quantity": 310,
      "approved_quantity": null,
      "kitchen_overflow_kg": 17.46
    },
    {
      "sku_id": "SKU-TOOR-DAL-1KG",
      "name": "Unpolished Toor Dal (1kg)",
      "category": "fmcg",
      "trailing_avg_Aiw": 22.0,
      "bottom_up_Eh": 0.0,
      "divergence_pct": 0.0,
      "base_demand": 22.0,
      "day_constant_c": 0.97,
      "month_factor_M": 1.0,
      "rejection_factor_R": 1.0,
      "reconciled_demand_D": 21.34,
      "initial_buffer_percentage": 25.0,
      "buffer_percentage_applied": 25.0,
      "order_up_to_S": 26.68,
      "sound_on_hand": 0.0,
      "raw_order": 26.68,
      "case_pack_size": 6,
      "moq": 12,
      "proposed_quantity": 30,
      "approved_quantity": null,
      "kitchen_overflow_kg": 0.0
    }
  ],
  "reviewed_by": null,
  "reviewed_at": null,
  "override_reason": null
}
```

Test 4: End-to-End Aging Exclusion Data Flow (`test_04_end_to_end_aging_exclusion_flow`)

Requests Sent:


```
{
  "step_1": {
    "method": "POST",
    "url": "/jobs/fresh-routing?run_sync=true"
  },
  "step_2": {
    "method": "POST",
    "url": "/jobs/store-pull?run_sync=true&delivery_date=2026-11-11"
  },
  "precondition": {
    "TEST-BATCH-PANEER-10H": "6 units @ 10h",
    "TEST-BATCH-PANEER-52H": "10 units @ 52h"
  }
}
```


Response Received (`HTTP 200`):

```
{
  "fresh_routing_routed_to_kitchen": [
    {
      "batch_id": "TEST-BATCH-PANEER-52H",
      "sku_id": "SKU-PANEER-FRESH",
      "name": "Artisanal Fresh Paneer (200g)",
      "category": "dairy",
      "quantity": 10.0,
      "age_metric": "52.0 hours elapsed",
      "status": "ROUTE_TO_KITCHEN",
      "action_notes": "Approaching freshness limit (52.0h > 48h); routed to Swadishtt kitchen."
    }
  ],
  "store_pull_paneer_item": {
    "sku_id": "SKU-PANEER-FRESH",
    "name": "Artisanal Fresh Paneer (200g)",
    "category": "dairy",
    "trailing_avg_Aiw": 18.0,
    "bottom_up_Eh": 0.0,
    "divergence_pct": 0.0,
    "base_demand": 18.0,
    "day_constant_c": 0.97,
    "month_factor_M": 1.0,
    "rejection_factor_R": 1.0,
    "reconciled_demand_D": 17.46,
    "initial_buffer_percentage": 10.0,
    "buffer_percentage_applied": 10.0,
    "order_up_to_S": 19.21,
    "sound_on_hand": 6.0,
    "raw_order": 13.21,
    "case_pack_size": 1,
    "moq": 5,
    "proposed_quantity": 14,
    "approved_quantity": null,
    "kitchen_overflow_kg": 1.75
  }
}
```


---

4. Code Updates & Bug Fix Log

During the migration from the in-memory mock repository to local PostgreSQL and execution of the edge-case suite, **6 bugs / architectural gaps** were discovered and fixed:

Bug 1: Doorstep Rejection Factor (`R(i)`) Suppressed Customer Demand

- **File:** `app/core/pull_theory.py` (Lines 129–148)
- **Issue Found:** The original code multiplied `D_next` by `R_i` (`D_next = base_demand * day_constant_c * festival_factor_F * month_factor_M * R_i`). When poor farm quality caused doorstep rejections (e.g., `R = 0.65`), the engine artificially shrank demand by 35% instead of maintaining true demand and alerting procurement.
- **Fix Applied:** Removed `* R_i` from `D_next` (`D_next = base_demand * day_constant_c * festival_factor_F * month_factor_M`), and added `ProcurementAlert` generation whenever `R_i < 0.95` to instruct procurement

to swap supplier/farm lots.

Bug 2: Kitchen Capacity (`K`) Trimming Stopped at a 5% Floor & Lacked Category Splits

- **File:** `app/core/pull_theory.py` (Lines 159–230) & `app/models/schemas.py` (Lines 18–22)
- **Issue Found:**
  1. `ProductCategory` used `HERBS_DAIRY` and `PACKAGED_GOODS_FMCG` instead of Phase 1 schema categories (`leafy_greens`, `dairy`, `base_veg`, `fmcg`).
  2. Buffer trimming stopped at a hardcoded `0.05` (5%) floor and `max_iterations = 25`, failing to bring massive overflow under `K_cap`.
  3. Only checked a single aggregate `K_cap` rather than category-specific `K` limits from `cdao_kitchen_capacity`.
- **Fix Applied:** Aligned `ProductCategory` enum to `leafy_greens`, `dairy`, `base_veg`, `fmcg`; wired `cdao_kitchen_capacity` category caps; increased `max_iterations = 200`; updated buffer trimming to reduce down to `0.0%` strictly in priority order `leafy_greens` → `dairy` → `base_veg`, satisfying both aggregate and per-category caps.

Bug 3: Standalone `/jobs/store-pull` Halved Packaged FMCG Demand When `E_h` Was Uninitialized

- **File:** `app/core/pull_theory.py` (Lines 111–127)
- **Issue Found:** When `/jobs/store-pull` ran before `/jobs/hce-nudge`, `latest_bottom_up_eh` in `cdao_system_config` was `{}`. `eh_map.get(s.sku_id, 0.0)` defaulted to `0.0` (100% divergence from `A_iw`), causing packaged goods (`fmcg`) to average `(A_iw + 0.0) / 2` and cut demand in half (e.g. `SKU-ATTA-5KG` dropped from `18.0` to `9.0`).
- **Fix Applied:** Guarded the HCE cross-check with `if eh_map and s.sku_id in eh_map:` so `A_iw` is preserved cleanly when HCE has not yet run.

Bug 4: Direct Reference to `db.batches` in FEFO Sorting Crashed Against PostgreSQL

- **File:** `app/core/no_expiry.py` (Lines 42, 50, 122)
- **Issue Found:** `sellable_items.sort(key=lambda x: db.batches[x.batch_id].manufacture_or_harvest_date)` bypassed `BaseRepository` and accessed an in-memory dictionary attribute not present on `PostgresRepository`.
- **Fix Applied:** Cached `batch_ref_dates[b.batch_id] = ref_date` while iterating over the batches returned by `await db.get_all_batches()`, then sorted `sellable_items` via `batch_ref_dates[x.batch_id]`.

Bug 5: UTC vs. IST (`+5.5h`) Timezone Skew Inflated Batch Harvest Ages

- **Files:** `sql/01_schema_ddl.sql` (Lines 33–36, 73, 81, 95, 101, 125), `app/core/no_expiry.py` (Lines 16–20, 39), `app/core/hce_engine.py` (Lines 17–20, 34), `app/core/pull_theory.py` (Line 66), `app/repository/postgres.py` (Line 534)
- **Issue Found:** The Docker PostgreSQL container runs in `UTC`, inserting UTC timestamps via SQL `NOW()` into `TIMESTAMP` (without timezone) columns, while host Python `datetime.now()` ran in local `IST (UTC+05:30)`. Subtracting naive UTC `harvest_date` from local IST `now` inflated every batch's age by `+5.5 hours` (e.g., a `52h` batch appeared as `57.5h`), risking premature kitchen routing for batches near `42.5h-48h`.

- **Fix Applied:** Changed all SQL timestamp columns to `TIMESTAMPTZ` and standardized elapsed time comparisons in Python on `datetime.now(timezone.utc)` with a `_to_utc()` helper. Verified in `test_04`: `52.0h` batch now measures accurately as `"52.0 hours elapsed"`.

Bug 6: `APScheduler` Global Instance Failed Across Multiple `pytest` Event Loops

- **File:** `app/scheduler.py` (Lines 10–51)
- **Issue Found:** Instantiating `scheduler = AsyncIOScheduler()` globally at import time bound it to the first event loop and broke across subsequent test teardown/startup lifespans.
- **Fix Applied:** Instantiated `AsyncIOScheduler()` dynamically inside `start_scheduler()` and reset `scheduler = None` in `shutdown_scheduler()`.